

Calibrating Multi-Agent Social Simulation and Simulation-Based GEO Prediction

A Research & Engineering Plan

Calibrating Multi-Agent Social Simulation to Predict AI Visibility of Product Descriptions

This document describes a three-tier research program. The same core engineering work supports all three tiers: a production-grade GEO prediction pipeline, a publishable paper at an information systems or financial NLP venue, and a path to a top-tier ML conference paper. You are expected to read this document, understand the full picture, and execute.

0. The Big Picture

The central question driving this project is:

How does the way a product is described determine whether an AI will recommend it?

This is the problem of Generative Engine Optimization (GEO) — the successor to SEO in the age of LLMs. When a user asks ChatGPT or Perplexity "which credit card should I get for travel?", the answer is not determined by a search index. It is determined by what the model has internalized about those products from the text it has seen.

The hypothesis of this project is that the path from product description to AI recommendation passes through social discourse. A product description that gets people talking — that generates authentic, varied, opinionated discussion — is more likely to eventually be represented in AI training corpora in a way that produces recommendations. We use MiroFish-style multi-agent simulation to model this social discourse process, and we calibrate that simulation against real-world Reddit data so that it has predictive validity.

1. Three Tiers of Ambition

This project has three levels. The engineering work is shared. The difference is how far you push the science.

Tier	Description	Status
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Tier 1	Top ML Conference (ICML / NeurIPS / ICLR) Title: "Calibrating LLM-based Multi-Agent Social Simulation to Real-World Dynamics" Multiple domains. Rigorous benchmark. Full theoretical framing.	Target
Tier 2	Engineering Pipeline A clean, documented, reproducible codebase implementing the full simulation-calibration-GEO pipeline. This is the base deliverable everything else depends on.	Required
Tier 3	Information Systems / Financial NLP Paper Title: "How Product Descriptions Shape AI Visibility: Evidence from Credit Card Recommendations" Single domain. Correlation-based claims. Safe exit.	Safety Net

Guiding Principle

Build Tier 2 first. Do it well. Tier 3 falls out almost automatically. Tier 1 requires extending Tier 2 to multiple domains and tightening the science. The order matters.

2. Why This Approach Is Novel

Existing GEO research treats the problem as a static text optimization task: take a product description, run it through an LLM, measure citation rate, repeat. This is essentially prompt engineering applied to marketing copy.

Our approach is fundamentally different. We model the social propagation mechanism that sits between a product description and an AI recommendation. The key insight is:

- AI recommendations are shaped by training corpora.
- Training corpora reflect what humans wrote and shared.
- What humans write and share is a social emergence process.
- Social emergence processes can be simulated with multi-agent systems.
- Therefore: the quality of a product description should be measurable by simulating its social propagation, not just by feeding it directly to an LLM.

The novelty is the middle step: using calibrated simulation as a proxy for the social dynamics that ultimately determine AI visibility. No existing GEO paper does this.

3. The Full Technical Pipeline

The pipeline has five stages. Stages 1-3 constitute the calibration system. Stages 4-5 constitute the GEO measurement system. Together they form the complete engineering deliverable.

Pipeline Overview

Stage 1: Data Collection (real Reddit discussions) Stage 2: Feature Extraction (what makes real discussion real) Stage 3: Iterative Calibration (align simulation to reality using a learned judger) Stage 4: GEO Measurement (simulate discussion, inject into context, query AI) Stage 5: Analysis and Reporting (which description features predict GEO score)

Stage 1: Data Collection

You need two things: real product descriptions and real human discussions of those products.

1a. Product Descriptions

- Select 30-50 real credit cards from publicly available sources: NerdWallet, Bankrate, Credit Karma, and each bank's official product page.
- For each card, collect the official marketing description as it appears on the bank's website. This is your input variable.
- Record key attributes: card name, issuing bank, reward structure, annual fee, target customer segment.

1b. Reddit Discussion Data

- Target subreddits: r/creditcards, r/personalfinance, r/churning.
- For each card in your set, query Reddit for posts and comments mentioning that card. Use PRAW (Python Reddit API Wrapper) with a free academic account.
- Collect at minimum 50 posts/comments per card where that card is the primary subject of discussion.
- Store: post text, comment text, upvote counts, comment depth, timestamp.
- This dataset is your ground truth for what real human discussion of a financial product looks like.

Data Split

Split your 30-50 cards into training (70%) and test (30%) sets before you do anything else. The test set cards must never influence calibration. This split is the foundation of your scientific claim.

Stage 2: Feature Extraction

The calibration system needs to know what makes a piece of simulated text look like real Reddit discussion. You need to operationalize this.

Feature Categories to Extract from Real Reddit Data

- **Semantic coverage:** Topic coverage:
 - What aspects of the card get discussed? (rewards, fees, customer service, approval odds, comparison to competitors)
 - Build a topic distribution for each card using LDA or BERTopic. This is your reference distribution.
- **Sentiment:** Sentiment and opinion structure:
 - Is the discussion uniformly positive, or does it contain genuine disagreement and nuance?
 - Real discussion tends to have mixed sentiment. Simulated discussion often skews one way.
- **Register:** Linguistic register:
 - Reddit has a distinctive voice: informal, first-person, experience-based ("I've had this card for two years and...").
 - Extract n-gram features that characterize real Reddit financial discussion.
- **Interaction:** Interaction patterns:
 - Do agents ask questions? Do they respond to each other? Do opinions shift?
 - These structural features of dialogue distinguish real social discourse from generated monologue.

Stage 3: Iterative Calibration

This is the core technical contribution. The goal is to make MiroFish's simulation of discussion about a credit card match the feature distribution of real Reddit discussion about that card.

3a. The Judger Model

Train a binary classifier: given a piece of text about a credit card, is this real Reddit discussion or MiroFish-generated simulation?

- Positive examples: real Reddit posts and comments from Stage 1.
- Negative examples: initial MiroFish output before calibration.
- Backbone: FinBERT or DeBERTa-base. Fine-tune on your training set cards only.
- This judger is your calibration signal. A simulation is "good enough" when the judger cannot reliably distinguish it from real discussion.

3b. The Calibration Loop

The loop runs as follows:

1. Start with a few-shot prompt: include 5-10 real Reddit comments about the card's category as exemplars in the MiroFish agent persona.
2. Run MiroFish simulation for the product description. Generate a discussion of 50-100 agent turns.
3. Pass the generated discussion through the judger. Each turn gets a probability score: $P(\text{real})$.
4. Turns with $P(\text{real})$ above threshold (e.g., 0.6) are kept as high-quality exemplars and added to the exemplar pool.
5. Turns below threshold are discarded.

6. In the next iteration, the expanded exemplar pool is used as few-shot context for MiroFish.
7. Repeat until convergence: the average P(real) of generated turns stops improving, or you have run N iterations.

Why This Works

This is Quality-Filtered Self-Play. Each iteration, the exemplar pool gets better because only realistic outputs survive. MiroFish then conditions on more realistic exemplars and produces more realistic output. The judge provides a stable, cheap calibration signal without requiring any gradient computation.

3c. Convergence Criteria

- Primary: Average judge score of generated turns exceeds 0.70 on held-out real examples.
- Secondary: Topic distribution KL-divergence between simulated and real discussion drops below threshold.
- Practical: If no improvement after 5 iterations, stop and report.

Stage 4: GEO Score Measurement

Once you have a calibrated simulation, you can measure GEO score for any product description.

4a. Generating Simulated Discussion

- For each card in your test set, run the calibrated MiroFish pipeline to generate a full simulated discussion (100-200 agent turns).
- Also run the uncalibrated pipeline as a baseline.

4b. Injecting Into AI Context

You cannot retrain ChatGPT. Instead, you use a RAG-style simulation of what AI would know:

8. Take the simulated discussion as a document corpus.
9. Build a retrieval index over this corpus.
10. Construct a prompt: "Based on the following community discussions, which credit card would you recommend for [user profile]? [retrieved passages]".
11. Query GPT-4 and Perplexity with multiple user profiles (frequent traveler, student, cash back maximizer, small business owner).
12. Record: was this card mentioned? Was it recommended? Was it the top recommendation?

4c. GEO Score Definition

GEO score for a card = weighted average of recommendation rate across user profiles. Weight by how well the user profile matches the card's stated target customer. This gives you a single number per card that is your dependent variable.

Stage 5: Analysis

Now you have, for each card in the test set:

- The original product description (text features extractable).
- The GEO score from calibrated simulation.
- The GEO score from uncalibrated simulation (baseline).
- Optionally: the real GEO score measured by directly querying AI without injected context.

Primary Analysis: Which Description Features Predict GEO Score?

- Extract features from the product description: length, specificity of benefit claims, number of use cases mentioned, presence of comparison language, emotional vs. functional framing, etc.
- Run regression analysis: which features correlate with GEO score?
- Report Spearman correlation between simulated GEO score and real GEO score on test set.

Secondary Analysis: Does Calibration Help?

- Compare correlation between simulated and real GEO scores for calibrated vs. uncalibrated pipeline.
 - This is your main ablation and directly validates the calibration contribution.
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4. Tier 3: The Financial Paper (Safe Exit)

If you complete Stage 1-5 cleanly, this paper writes itself. The story is:

Paper Claim

We propose a simulation-based framework for measuring the AI visibility of financial product descriptions. Using calibrated multi-agent social simulation, we show that simulated discourse quality predicts real AI recommendation rates (Spearman rho = X.XX, $p < 0.01$) on a benchmark of N credit card products. We further identify the linguistic and structural features of product descriptions that most strongly determine GEO performance.

Paper Structure

Section	Content
Introduction	Why GEO matters for financial products. Why social simulation is the right lens. Why credit cards are the right domain (non-commodity, AI-recommendable, rich public discussion).
Related Work	GEO literature (Aggarwal et al. 2023 and subsequent). Multi-agent simulation (MiroFish, OASIS). Financial NLP. LLM-as-judge.
Method	The five-stage pipeline. Calibration loop. GEO score definition. This section is already written if your code is clean.
Experiments	Dataset statistics. Calibration results (judge accuracy, convergence curves). Main result: correlation between simulated and real GEO. Ablations: calibrated vs. uncalibrated, different user profiles.
Analysis	Which description features matter? Qualitative examples of high vs. low GEO descriptions.
Conclusion	Implications for financial content strategy in the AI era. Limitations. Future work (Tier 1 setup).

Target Venues

- ICIS (International Conference on Information Systems) — December deadline cycle.
 - WSDM (Web Search and Data Mining) — November deadline, accepts applied track papers.
 - FinNLP workshop at IJCAI or EMNLP — shorter paper, faster turnaround.
 - AAAI ICWSM — social simulation angle fits well here.
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5. Tier 1: The Top Conference Paper (The Goal)

The Tier 3 paper proves the concept in one domain. The Tier 1 paper proves it is a general phenomenon. The question becomes: is social simulation a valid proxy for real-world propagation dynamics, across domains?

What Changes from Tier 3 to Tier 1

- **Domains:** Multiple domains:
 - Credit cards (financial products — already done in Tier 3).
 - Consumer electronics (e.g., smartphones, laptops).
 - Software tools (e.g., productivity apps, developer tools).
 - At minimum 3 domains, each with its own Reddit community baseline.
- **Scale:** Larger scale:
 - 100+ products per domain. More robust statistics.
- **Theory:** Stronger theoretical framing:
 - Connect to social contagion theory and information cascade literature.
 - Formalize the calibration procedure mathematically.
- **Benchmark:** Richer benchmark:
 - Release the dataset and evaluation protocol publicly so others can compare.
 - This is what makes it a community resource, not just a one-off study.
- **Claims:** Tighter claims:
 - The Tier 3 paper reports correlations. The Tier 1 paper explains why the correlations exist, with ablations that isolate the causal mechanism.

The Tier 1 Title and Claim

"Calibrating LLM-based Multi-Agent Social Simulation to Real-World Dynamics: A Benchmark and Framework"

Core claim: Quality-Filtered Self-Play with a feature-based judger is a principled and generalizable method for aligning multi-agent LLM simulation with real human social dynamics. We provide the first multi-domain benchmark for evaluating simulation fidelity, and demonstrate downstream utility in GEO prediction.

6. Known Open Questions

These are real problems. You are not expected to solve all of them for Tier 3. For Tier 1, you will need answers to most of them.

Question	Detail
Judger Generalization	The judger is trained on real vs. simulated text for the training set cards. How well does it generalize to test set cards? If it overfits to card-specific vocabulary, calibration quality on test set will be artificially inflated. Measure this with cross-card generalization experiments.
Simulation Stochasticity	MiroFish is non-deterministic. Two runs with the same input produce different discussions. GEO scores will therefore have variance. You need to run multiple simulations per card and report confidence intervals, not point estimates.
RAG vs. True Training	The GEO measurement step uses RAG injection, not actual model retraining. This is a simplification. The claim is that RAG injection is a reasonable proxy for training-time exposure. This needs to be stated clearly as a limitation and partially validated by comparing RAG-based GEO scores to direct LLM query results on a subset.
Reddit as Ground Truth	Reddit discussion is not the only path from product description to AI training corpus. There are review sites, news articles, forum posts. Using Reddit as the sole calibration target is a simplification. Acknowledge this and argue that Reddit is a representative and measurable proxy.
Judger Backbone Choice	FinBERT may not be the best choice. DeBERTa-base is more general and often outperforms domain-specific models on classification. Run both and report. For Tier 1, this becomes an ablation.

7. Suggested Milestone Structure

This is a suggestion, not a requirement. Adjust based on your own working style.

Milestone	Definition of Done
Milestone 1	Data collection complete. 30-50 credit cards with product descriptions and Reddit discussion corpora. Train/test split finalized and locked. Do not touch the test set after this point.
Milestone 2	Feature extraction pipeline working. You can take a Reddit post and produce a feature vector. You can compute topic

	distribution for a card's discussion corpus. Judger training data is ready.
Milestone 3	Judger trained and evaluated. Binary classifier distinguishes real from initial simulated discussion with accuracy above 0.75 on held-out training set cards. If accuracy is below 0.75, investigate why before proceeding.
Milestone 4	Calibration loop running end-to-end. For at least 5 training set cards, you have run the full iterative calibration and can show convergence curves. Judger score improves across iterations.
Milestone 5	GEO measurement pipeline working. For at least 5 test set cards, you have simulated GEO scores and real GEO scores. Compute initial correlation. If correlation is near zero, diagnose before scaling up.
Milestone 6	Full results on all test set cards. Main tables and figures ready. Tier 3 paper is now writable.
Milestone 7 (Tier 1 only)	Second and third domain data collected. Pipeline adapted and re-run. Multi-domain benchmark assembled. Paper extended.

8. What Success Looks Like

Minimum Success (Tier 3 publishable)

- Spearman rho > 0.40 between calibrated simulation GEO score and real GEO score on test set, $p < 0.05$.
- Calibrated pipeline outperforms uncalibrated pipeline on this metric (ablation holds).
- At least 3 interpretable description features identified that significantly predict GEO score.
- Clean, reproducible codebase with documentation.

Strong Success (Tier 1 competitive)

- Rho > 0.55 on credit card domain.
- Results replicated on 2 additional domains.
- Judger generalization ablation is clean: calibration quality on test set matches training set.
- Public benchmark released with leaderboard potential.

What Failure Looks Like (and What to Do)

- Rho near zero: the simulation is not tracking real discussion dynamics. First check: is the judger actually working? If judger accuracy is high but GEO correlation is low, the problem may be in the RAG injection step or the GEO score definition. Debug methodically.

- Judger accuracy below 0.65: the features you chose are not discriminative enough. Try different backbone, add more training data, or revisit the feature extraction.
 - Calibration loop does not converge: the exemplar pool may not be diverse enough. Try different initialization or looser acceptance threshold.
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9. Key Resources

Codebases

- MiroFish: github.com/666ghj/MiroFish — primary simulation engine. Read the README fully before touching the code.
- MiroFish-Offline: github.com/nikmcfly/MiroFish-Offline — fully local version using Ollama and Neo4j. Use this if API costs are a concern.
- OASIS: the underlying multi-agent social simulation framework used by MiroFish. Understanding this helps you customize agent behavior.
- PRAW: Python Reddit API Wrapper. Free for academic use. reddit.com/dev/api
- FinBERT: huggingface.co/ProsusAI/finbert — financial domain BERT.
- DeBERTa-base: huggingface.co/microsoft/deberta-base — strong general classifier.

Literature to Read Before Starting

- Aggarwal et al. (2023). GEO: Generative Engine Optimization. The foundational paper defining the field.
- MiroFish documentation and demo videos — especially the public opinion simulation demo.
- OASIS paper from CAMEL-AI team — understand the simulation architecture.
- LLM-as-Judge literature: Zheng et al. (2023) "Judging LLM-as-a-Judge with MT-Bench" — the conceptual basis for using LLMs as quality evaluators.
- Guo et al. (2023) "EPC: Evaluating the Performance of LLM-based Agents" — relevant for thinking about agent calibration.

Data Sources

- Reddit: praw.readthedocs.io — register at reddit.com/prefs/apps for academic credentials.
 - NerdWallet, Bankrate, Credit Karma — scrape product descriptions directly. These are public pages.
 - Bank official product pages: Chase, Citi, Capital One, Amex, US Bank, Bank of America — all public.
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